



CHEMISTRY Department
SIMHADRI



Ref:SMTTPP/Chemistry/2022-23/LMI/828064

Date:08/02/2023(10:01:39)

Subject:LMI on Recommendations of Chemistry Heads Meet-2022

LMI doc no.LMI/OIN/OPS/CHEM/047 REVISION 0 , " **LMI on Recommendations of Chemistry Heads Meet-2022** " has been made in line with COS-ISO-00-OIN/OPS/CHEM/047 Rev. No.: 0 Dec. 2022. The LMI covers the details about various technical presentations held during workshop and recommendations by COS - Chemistry along with final responsibilities.

This LMI is being forwarded to all concerned departments for information sharing and necessary action at their end if any recommendation is applicable.

Submitted for kind perusal and approval please.

Submitted by:

सुनीता गुलीपिल्ली - प्रबंधक

SUNEETA GULIPILLI - 101282 (MANAGER ,CHEMISTRY) ,Simhadri
(SUNEETAGULIPILLI@NTPC.CO.IN, 9493547082)

On:08/02/2023(10:01:39) **Ref:**SMTTPP/Chemistry/2022-23/LMI/828064

Submitted by:

दीपशिखा वर्मा - उप महाप्रबंधक (रसायन)

Deepshikha Verma - 007773 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(DEEPSHIKHAVERMA@NTPC.CO.IN, 9650995244) ,Simhadri

On:08/02/2023(10:29:52) **Ref:**SMTTP/Chemistry/2022-23/LMI/828064

May kindly be approved

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:09/02/2023(09:39:45) **Ref:**SMTTP/Chemistry/2022-23/LMI/828064

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:09/02/2023(17:00:12) **Ref:**SMTTP/Chemistry/2022-23/LMI/828064

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:09/02/2023(18:05:39) **Ref:**SMTPP/Chemistry/2022-23/LMI/828064

May please be approved.

Submitted by:

ASIS KUMAR MUKHOPADHYAY - 004586 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF
SITE)

(AKMUKHOPADHYAY@NTPC.CO.IN, 9434038935) ,Simhadri

On:10/02/2023(16:17:54) **Ref:**SMTPP/Chemistry/2022-23/LMI/828064

Review

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:13/02/2023(11:24:10) **Ref:**SMTTPP/Chemistry/2022-23/LMI/828064

Responsibilities are inline.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(RRSATPATHY@NTPC.CO.IN, 9437021584) ,Simhadri

On:17/02/2023(16:00:21) **Ref:**SMTTPP/Chemistry/2022-23/LMI/828064

Responsibilities are inline.

May be approved.

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:18/02/2023(10:00:05) **Ref:**SMTTP/Chemistry/2022-23/LMI/828064

Submitted by:

विश्व मोहन चौधरी - महाप्रबंधक

Vishwa Mohan Choudhary - 003932 (GENERAL MANAGER ,FUEL MANAGEMENT)
(VMCHOUDHURY@NTPC.CO.IN, 7588691232) ,Simhadri

On:18/02/2023(17:17:30) **Ref:**SMTTP/Chemistry/2022-23/LMI/828064

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:21/02/2023(11:08:54) **Ref:**SMTTPP/Chemistry/2022-23/LMI/828064

May please be approved.

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:21/02/2023(17:49:43) **Ref:**SMTTPP/Chemistry/2022-23/LMI/828064

Approved & Submitted by:

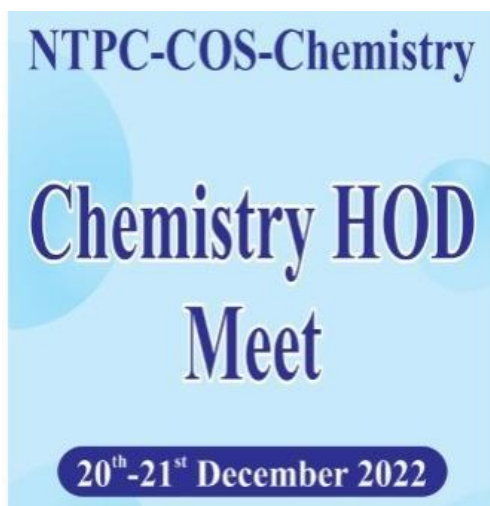
संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

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LMI- OIN/OPS/CHEM/047 REVISION 0

**LMI on Recommendations of Chemistry
Heads Meet-2022**



Approved for implementation by GGM (Simhadri)

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1.0 SUMMARY :

1.1 Superseded Document	Nil	
1.2 Scope	NTPC SIMHADRI	
	Review Responsibility	O&M - CHEMISTRY
1.3 Review	Review Frequency	NA – One time activity.
	Review Approval	HOP - SIMHADRI

2.0 INTRODUCTION :

- i) Chemistry Heads Meet-2022 of all NTPC stations, JVs and Subsidiaries was organized from 20.12.2022 to 21.12.2022 at Raipur and was attended by the Chemistry HODs/ representatives from all NTPC stations, JVs and Subsidiaries.
- ii) The meet was inaugurated by RED (WR-II) & Executive Director (Operation Services) Sh A.K.Tripathy in presence of other senior dignitaries viz. Sh C. Sivakumar, ED (USSC) and Sh V. Chandrasekharan, AGM(COS-Chemistry).
- iii) AGM(COS-Chemistry) in his inaugural address extended a warm welcome to all the participants from all the NTPC stations, JVs and subsidiaries. He briefly explained the genesis of the meet and gave a noting on the content of the meet. In his speech he raised concern over the recent increase in Chemistry related failures and issues in boiler and turbine liabilities of NTPC stations. Further, he added that in this era of load flexibilization utmost care is to be taken for maintaining Cycle Chemistry Parameters to prevent any FO/PL due to Chemistry issues.
- iv) ED (USSC) appreciated the COS team for organising this meet and felt that Chemistry function plays a very crucial role in each activity of a power plant. He stressed that Chemistry group should work in line with the organisational goal of safe, efficient, and reliable power generation. He deliberated upon the recent & new initiatives undertaken by USSC through Rate contract of bulk chemicals and CWT chemicals.
- v) He also stressed the fact that how smooth coordination with vendors at site level, correct projected requirement quantity of chemicals and addressing payment issues to vendors can ensure timely supply of chemicals at Site. He assured for complete support to Chemistry function from USSC regarding all critical issues faced by the stations.
- vi) The event was formally launched by RED (WR-II) & Executive Director (Operation Services). In his inaugural speech, he emphasised the relevance of the vital role of Chemistry parameters in the operation paradigm of our units.
- vii) He appreciated the session contents of the HOD meet and felt that interactive session

with all stakeholders' departments pertaining to Chemistry function would benefit the routine activities of Chemistry department.

- viii) He further contemplated upon the fact that in the future NTPC is going to play a decisive role in the load balance scenario of the country and our units should be able to run as per our wish to meet those challenges.
- ix) He stressed upon the fact with availability of healthy CPU system, no Partial Loss should be reported for units coming after outage. He specially mentioned that the older units should give special attention to routine OH jobs including structural healthiness of WTP equipments and a rolling plan is to be maintained for repair schedule.
- x) Units reporting repeated SWAS parameter aberration should try to resolve the issues in a time-bound manner and non-availability of critical online instrumentation is to be discussed in Site DPM. He urged that the units experiencing turbine deposits should carry-out proper root cause analysis with action plan to prevent any recurrence during running of the units.
- xi) Further, he said that during the meet all engineering issues related to chemistry should be addressed in the upcoming Supercritical units.
- xii) He mentioned that with the advent of permanent Chlorine Dioxide package, due diligence should be given towards the benchmarking of Cost optimisation. He spoke in detail about the importance of Coal Quality monitoring for a station and its impact on the organisation bottom line performance. He made it a point that all our coal testing laboratories should be NABL accredited, and COS Chemistry should support the sites for compliance of the same.
- xiii) Finally, he suggested that the recommendations of the meet should be finalised in the concluding session of the event.
- xiv) During the inaugural session, Chemistry Benchmarking Score (CBS) dashboard was launched by RED (WR-II) & Executive Director (Operation Services) Sh A.K. Tripathy. CBS is a part of "Advanced Data Analytics and Visualization Platform" of COS-Digitization group and shall be rolled out for all stations by Jan-2023.
- xv) At the end of inaugural session, Sr Manager (COS-Chemistry) during his "vote of thanks" address, expressed heartfelt thanks and sincere gratitude to the senior dignitaries and all the participants from NTPC stations, JVs and subsidiaries for attending the meet and assured for complete support to Site Chemistry in resolving all issues by working together.
- xvi) He thanked all COS functions, NETRA & PE(WS) for their scheduled interactions in the meet and contribution of COS(DI) and M/S Fluent grid for all support in introducing the Chemistry benchmarking score. He specially thanked HR and IT team for providing all logistics, infrastructure, and cooperation in organising the meet.

3.0 RECORDS OF REVISION - Nil

4.0 Presentations held during the Workshop:

After the inaugural session, the meet started with the introductory presentation by COS-Chemistry team. An overview of the Chemistry performance for the last FY with a focus on Chemistry-induced PL/FO data and critical issues of Chemistry functions were deliberated in the presentation. The team also put forward the new guidelines as per the revised TC documents and its importance. This two day meet was attended by 42 participants from OS, NTPC Stations, JVs and subsidiaries, CC-Engg-PE (WS) and CC-NETRA. Following Presentations and interactive sessions were held during the meet:

- i) Presentation on New Engineering Initiatives, Design issues related to CPU, DM &RO, STP- water recycling and ZLD by PE (WS).
- ii) Presentation on Chemistry-related BTL and RCA, preservation infrastructure issues, oil condition monitoring, mill isokinetic sampling, related TC documents by COS – Boiler.
- iii) Turbine, Condenser and CT efficiency issues by COS-CENPEEP
- iv) Interaction with COS-BOP group (presentation on DM-PT -CPU liabilities, CW System issues).
- v) Presentation by EPRI team on recent advancements in Steam -water Cycle Chemistry parameter management, Turbine deposition mitigation measures and discussion on case studies.
- vi) Presentation on CWT formulation, CWT chemical specifications and quality checks, Case studies, condenser tube/turbine deposit analysis, Fire-water system corrosion & its mitigation, Oil condition monitoring by CC-NETRA.
- vii) Interaction with COS-ST group (presentation on partial loss due to turbine deposit issues, CTL,OH-SOW, CPU liabilities, oil condition monitoring).
- viii) Interaction with CC-FM group on coal and GCV related station issues.
- ix) Technical Presentation by M/S GE group on role of Chemistry in FGD system.
- x) Interaction of COS-Operation on expectation of Operation group from Chemistry under newCERC regulations.
- xi) Interaction with CPG-1 - PPG- Group on bulk chemical & instrument/spares procurement and planning.
- xii) Based on all presentations and discussions thereof, recommendations of the meet were finalized in the concluding session on the last day.

5. Recommendations

The final recommendations along with responsibilities are summarized in the following table:

Sl. No.	Recommendation	Responsibility
1.	Chemistry benchmarking score trial run is to be rolled out from Jan'23 for all stations.	COS-Chemistry/All Sites
2.	Based on learning of DSTPS U#1 resin leakage issue, auto isolation of all service-in CPUs to be ensured whenever all CEPs are 'Not ON' for 5 seconds.	COS C&I
3.	Procurement of PPEs for Chemical Safety is to be clubbed with bulk procurement undertaken by CC-Safety and IS standards for PPEs is to be finalized in consultation with CC-safety.	COS-Chemistry/CC-Safety
4.	To improve the rubber lining quality, quality plan for testing of rubber for inspection procedure is to be revised. Guidelines to be issued for rubber line testing of MSRL vessels.	COS-BOP
5.	Outage period for rubber lining activity is to be framed for inspection of all storage tanks including Chemical Storage tank/DMW storage tank for safe operation.	COS-BOP
6.	In ageing units, MSRL pipelines for acid handling may be replaced with CPVC line for trouble free operation.	COS-BOP/Site MM-OFS
7.	Quality of dry instrument air is to be ensured for dry air preservation of Boiler & auxiliaries during RSD. Stations having issues in Dry air quality should intimate COS - Chemistry.	Site Chemistry/COS Chemistry
8.	In drum units, Feed water DO and pH to be maintained in recommended limit as per OD .	Site Chemistry of all drum units.
9a.	Sodium : Phosphate ratio to be monitored on regular basis in drum units, preferably once/day during part load, to identify phosphate hide-out. In drum units, diluted sodium hydroxide can be dosed (below 160 kg/cm ²) where hide-out problem is reported. (Note: NaOH dosing to be done only when drum pressure is low).	Site Chemistry.
9b.	As per EPRI, mechanical carryover is mainly responsible for deposition in turbines in drum type units. It is recommended to check mechanical carryover when unit load is changing (ramp-up/ down).	Site Chemistry
9c.	In drum units, Sodium in saturated steam also to be monitored on continuous basis.	Site Chemistry of all drum units.
9d.	In drum units where HP Turbine deposit is observed (Simhadri-I,II, TSTPS Stg-II, FSTPS-II,III, RSTPS Stg-II,III, VSTPS-II), one additional sodium analyser is to be installed for continuous monitoring of sodium in SS & MS samples to cross-check the mechanical carry-over.	COS C&I

10.	For external testing of CW Chemicals, one more CSIR Lab is to be identified for CWT Chemicals testing.	CC-NETRA & COS-Chemistry
11.	Review frequency of CW treatment programme by CC-NETRA for all stations is to be done in every 2 years.	Site Chemistry & CC-NETRA
12.	To prevent any scaling issues in condenser, operating pH is to be maintained around 8.0 for safe operation of CW system in Closed Cycle operation.	Site Chemistry
13.	Reply for PI alerts of SWAS parameter is to come from stations within 1 hour of receiving the SMS alert.	Site Chemistry
14.	Stations to identify Units having liabilities of high attemperation and the same is to be communicated to COS for inclusion in Turbine OH SOW of respective units.	Site Chemistry & COS-Chemistry.
15.	Sampling issues of DHA for turbine preservation is to be taken up with Site & OS ST.	Site Chemistry/TMD and COS-ST
16.	For Units where CPU SV downstream line is MSRL, MOC of CPU service vessel downstream is to be changed from MSRL to CS.	Site Chemistry/TMD of respective stations
17.	Site where FGD is installed is to carry out the detailed quality assessment of GYPSUM as per the Specimen sheet of GE as a trial study.	Site Chemistry of respective stations
18.	Appeal to be made to Coal Controller for "Regrading of Mines" in case of consistent grade slippage over 03 consecutive months, between NTPC part and TPS part sample, when more than 65 Kcal/kg GCV difference and change of grade is observed. (Latest MOM with Coal Controller is to be taken as reference)	Site Chemistry/FM
19.	Monthly Communication is to be given to Coal Subsidiary/CIL in case of Grade Slippage/Enabling Condition.	Site Chemistry/FM
20.	Taking up issues related to Third Party Sampling in Executive Committee, Implementation Committee, Apex Committee and JS Level Committee	Site Chemistry/FM
21.	Timely entry of GCV data in SAP and delay analysis data in excel sheet is to be done.	Site Chemistry.
22.	Enabling condition on monthly basis is to be updated as per CC-FM guidelines.	Site Chemistry/FM

6. List of participants

Sl No	Station	Name (Sh/Smt)	Designation	Department
NTPC Station/JVs/Subsidiaries				
1	Anta	Arun Singh	Sr Manager	Chemistry
2	Barauni	Rajib Mukhopadhyay	AGM	Chemistry
3	Barh	Amit Maurya	Manager	Chemistry
4	Bongaigaon	Hiranmoy Misra	Sr Manager	Chemistry
5	BRBCL	S.K. Maurya	AGM	Chemistry
6	Dadri	R.P.Das	AGM	Chemistry
7	Darlipalli	Biswabharati Ghadei	Sr Manager	Chemistry
8	Farakka	Gautam Kundu	AGM	Chemistry
9	Gadarwara	Omkar Singh	DGM	Chemistry
10	Jhabua	Umesh D Gawade	Sr Manager	Chemistry
11	APCPL Jhajhar	Sumit Garg	Dy Manager	Chemistry
12	Kahalgaon	Bikash Chakraborty	AGM	Chemistry
13	Kanti	Santanu Jana	DGM	Chemistry
14	Kawas	Parmar Vijay Kumar Ratilal	Sr Manager	Chemistry
15	Khargone	Vikash Pandey	DGM	Chemistry
16	Korba	Vijay Garg	AGM	Chemistry
17	Kudgi	Anjan Pramanik	DGM	Chemistry
18	Lara	Sudipto Saha	DGM	Chemistry
19	Mouda	Priya	Sr Manager	Chemistry
20	Meja	Dilip Kumar	Sr Manager	Chemistry
21	Nabinagar	A.K. Sinha	AGM	Chemistry
22	NSPCL, Bhillai	O.P. Gondule	Sr Manager	Chemistry
23	Patratu	J. Panda	Manager	Chemistry
24	Ramagundam	K. Indumathi	Sr Manager	Chemistry
25	Rihand	S.K Mohanty	AGM	Chemistry
26	Simhadri	Suneetha. G	Manager	Chemistry
27	Singrauli	Vikesh Sharma	Sr Manager	Chemistry
28	Sipat	J. Chakraborty	AGM	Chemistry
29	Solapur	S.D. Kadam	Sr Manager	Chemistry
30	Talcher Kaniha	Vinod Ratan	AGM	Chemistry
31	Tanda	A.K. Singh	DGM	Chemistry
32	Unchahar	Radhesh Kumar	AGM	Chemistry
33	Vallur	R.Rohith	Sr Manager	Chemistry
34	Vindhyachal	H.N Pandey	AGM	Chemistry
35	Dulanga	Manas Ranjan Das	Sr Manager	Chemistry
36	BIPCL	K. K. Sharma	AGM	Chemistry
37	CC-NETRA	Kiran Diwakar	Manager	Chemistry
38	CC-Engg	Geet Arora	Sr Manager	PE(WS)
39	CC -OS	V. Chandrasekharan	AGM	Chemistry

40	CC -OS	Bidisha Sen	DGM	Chemistry
41	CC -OS	Kajal Acharyya	Sr Manager	Chemistry
42	CC -OS	Durgesh Lohiya	Sr Manager	Chemistry

